

Karthik Thyagarajan

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EDUCATION

Purdue University

West Lafayette, Indiana

B.S. of Computer Science & Artificial Intelligence (Double Major), GPA: 3.93

May 2027

- **Coursework:** Data Structures & Algorithms, Computer Architecture, Programming in C, Competitive Programming, OOP

EXPERIENCE

AI Research Intern

May 2026 – Aug 2026

Samsung Research America

Mountain View, CA

- Built an on-device signal router for always-on ambient AI on Bixby Edge AI, filtering device signals on-phone at 95% suppression and 90% intent-routing accuracy to reserve frontier models for events that need them.
- Co-architected Samsung's always-on cross-device agent orchestrator from scratch, owning half of a 25-module Kotlin runtime that ships on XR glasses and gives phone, glasses, and outside developers one platform to build agents on.
- "Built the runtime's safety and control plane": 'capability-based permissions, user confirmation flows, policy-gated tool dispatch, token and cost budgets, and PII-redacted audit logging, so every action an agent takes is gated, bounded, and recorded.'
- Built phone-control tooling that gives agents computer-use-style access to any app's UI under stale-screen and confirmation guards, plus native calendar, alarm, location, and health tools that reach apps with no exposed API.

Machine Learning Engineering Intern

Jun 2025 – May 2026

Peraton Labs

Silver Spring, MD

- Developed RL agent for IoT malware detection using PyTorch and graph neural networks, reducing exploration latency by 35% and increasing detection coverage by 25% across 500K+ daily device events.
- Built large-scale data analytics ETL pipelines and heterogeneous graph architecture with autoencoders to model device communication patterns, accelerating policy convergence by 40%.

Computer Vision Researcher

Feb 2025 – Aug 2025

Memories.ai

Remote

- Architected video memory framework for AR applications processing 10K+ streams using Python, Flask, and PostgreSQL, achieving 60% throughput improvement through frame sampling optimization.
- Published Python SDK on PyPI with 2K+ downloads, implementing async processing and REST API integration for video analysis workflows.

Data Engineering Intern

Aug 2024 – Dec 2024

AgRPA

West Lafayette, IN

- Built end-to-end weed detection pipeline processing 200GB+ drone imagery with Python and cloud storage, implementing parallel ETL workflows and optimized data partitioning for 40% faster analysis.
- Engineered segmentation models achieving 92% accuracy on 50K+ images through agile sprints, reducing herbicide usage by 60% and costs by \$150K annually.

PROJECTS & CLUBS

Repple | *Python (FastAPI), Supabase, PostgreSQL, OpenAI API, Pydantic*

Ongoing

- Built realtime multiplayer backend with synchronized workout sessions for iOS app with 200+ users.
- Developed a competitive gamification engine featuring custom ELO matchmaking algorithms, automated cron-based point aggregation, and generative AI workout plans that adapt to user performance metrics.

google-tools-mcp | *Python, MCP, Google APIs, OAuth, Pydantic*

Ongoing

- Built MIT-licensed MCP server unifying 169 tools across Drive, Docs, Sheets, Slides, Gmail, Calendar, and Forms behind a single OAuth flow with auto re-auth and dynamic scope handling.
- Designed read-before-edit guards, lazy authentication, and multi-account profile support to enable safe agent orchestration across the full Google Workspace from one context.

Caladrius (HackGT - 2nd Place) | *React Native, Python, LangGraph, AWS S3, REST API, Next.js*

Sep 2025

- Architected AI triage system with encrypted, zero-knowledge data transfer for HIPAA compliance.
- Built multi-agent diagnostic system with LangGraph achieving 85% triage accuracy on symptom analysis.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C/C++, JavaScript, SQL, React, Flask, Next.js, Android

Cloud & DevOps: AWS, S3, EC2, Lambda, GCP, Docker, Git, CI/CD, Linux, REST API

AI/ML: LLM (LangChain, RAG, CoT/Reasoning, RLHF), Agents, MCP, PyTorch, GAN, RL, Diffusion